

Qubit Electronics-Formula Sheet

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1 Qubit Introduction

Fundamental Qubit Representation and State Vectors

- $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$
- $|\alpha|^2 + |\beta|^2 = 1 = \langle\psi|\psi\rangle$
- $|\psi\rangle^\dagger = \langle\psi| = [\alpha^*, \beta^*]$
- $|\psi\rangle = c_1|000\rangle + c_2|001\rangle + \dots + c_8|111\rangle$

The Bloch Sphere Geometry

- $|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$
- $(0, 0, 1), (0, 0, -1), (1, 0, 0), (-1, 0, 0)$

Quantum Measurement and the Born Rule

- $P(|a\rangle) = |\langle\psi|a\rangle|^2$

Basis States in Different Representations

- $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
- $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
- $|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle), |-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$

Single-Qubit Unitary Operators

- $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- $\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
- $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
- $\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$

- $\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
- $R_x(\theta) = e^{-i\theta X/2}, R_y(\theta) = e^{-i\theta Y/2}, R_z(\theta) = e^{-i\theta Z/2}$

Two-Qubit Interactions: CNOT Gate

- $|A\rangle|B\rangle \rightarrow |A\rangle|A \oplus B\rangle$
- $\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

Density Matrix Formalism

- $\rho = |\psi\rangle\langle\psi|$
- $\hat{\rho} \equiv \sum_i p_i |\Psi_i\rangle\langle\Psi_i|$
- $\text{Tr}(\rho^2) = 1$ if pure state; < 1 if mixed state

Qubit Performance Metrics

- $\frac{1}{T_2} = \frac{1}{T_\phi} + \frac{1}{2T_1}$
- $N_{\text{gates}} \sim \frac{T_2}{t_{\text{gate}}}$

2 Superconducting Qubits

Fundamental Superconductivity and Cooper Pairs

- $\psi = \psi_0 \exp \left[-j \frac{1}{\hbar} (Et - \mathbf{p} \cdot \mathbf{x}) \right]$
- $\rho = \psi \cdot \psi^* = |\psi_0|^2$
- $B = 0$

Josephson Junction Dynamics

- $\dot{\phi} = \varphi_2 - \varphi_1$
- $I = I_C \sin \phi$
- $V = \frac{\Phi_0}{2\pi} \dot{\phi}, \Phi_0 = \frac{h}{2e} = \frac{2\pi\hbar}{2e}$
- $L = \frac{\Phi_0}{2\pi I_C \cos \phi}, L_j = \frac{\Phi_0}{2\pi I_C}$
- $E_j(1 - \cos \phi)$

LC Circuit Quantization

- $H = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$
- $\omega_{01} = \frac{1}{\sqrt{LC}}$
- $[\hat{\Phi}, \hat{Q}] = i\hbar, [\hat{a}, \hat{a}^\dagger] = 1$

Transmon Qubit Parameters

- $L(\Phi) = \frac{\Phi_0}{2\pi I_C \cos(2\pi\Phi/\Phi_0)}$
- $E_{01} = \hbar\omega_{01} \approx 10 \text{ GHz}$
- $E_{01} \neq E_{12}$
- $k_B T \ll \hbar\omega_{01}$
- $T_\pi \approx 4\text{--}20 \text{ ns}$
- $T_1, T_2 \sim 100\text{--}1000 \mu\text{s}$

3 Circuit QED

Fundamental Light-Matter Interaction

- $H = -\vec{d} \cdot \vec{E}(t)$
- $\Omega_R = \frac{dE_0}{\hbar}$

Resonator Physics

- $2L = m\lambda_m$
- $\Delta f = \frac{c}{2L}$
- $\omega_r = \frac{1}{\sqrt{L_r C_r}}$
- $Q = \frac{\omega_r}{\kappa}$

Jaynes-Cummings Model

- $H = \hbar\omega_r \hat{a}^\dagger \hat{a} + \frac{\hbar\omega_{01}}{2} \hat{\sigma}_z + \hbar g(\hat{a} \hat{\sigma}_+ + \hat{a}^\dagger \hat{\sigma}_-)$
- $H_{JC} = \hbar\omega_r \hat{a}^\dagger \hat{a} + \frac{\hbar\omega_{01}}{2} \hat{\sigma}_z + \hbar g(\hat{a} \hat{\sigma}_+ + \hat{a}^\dagger \hat{\sigma}_-)$

Coupling Strength

- $g = \omega_r \frac{C_g}{C} \left(\frac{E_j}{2E_c} \right)^{1/4} \sqrt{\frac{\pi Z_r}{R_k}}$
- $g/2\pi \approx 100 \text{ MHz to } 1 \text{ GHz}$

Dispersive Regime

- $|\Delta| = |\omega_{01} - \omega_r| \gg g$
- $\chi = \frac{g^2}{\Delta}$
- $H \approx (\omega_r + \chi \hat{\sigma}_z) \hat{a}^\dagger \hat{a} + \frac{\omega_{01}}{2} \hat{\sigma}_z$
- $\omega_r^{(0)} = \omega_r - \chi$
- $\omega_r^{(1)} = \omega_r + \chi$
- $\propto \chi \hat{\sigma}_z t$

Circuit QED Implementation

- $\sim 10^{-3} \lambda$
- $E_0 \approx 0.2 \text{ V/m}$
- $\approx 2 \times 10^4$

System Hamiltonian Hierarchy

- $H = \frac{Q_r^2}{2C_r} + \frac{\Phi_r^2}{2L_r} + 4E_c \hat{n}^2 - E_j \cos \hat{\phi} - g(\hat{b}^\dagger - \hat{b})(\hat{a}^\dagger - \hat{a})$
- $H = \omega_r \hat{a}^\dagger \hat{a} + \frac{\omega_{01}}{2} \hat{\sigma}_z - g(\hat{\sigma}_+ + \hat{\sigma}_-)(\hat{a}^\dagger + \hat{a})$
- $H = \omega_r \hat{a}^\dagger \hat{a} + \frac{\omega_{01}}{2} \hat{\sigma}_z + g(\hat{a}^\dagger \hat{\sigma}_- + \hat{a} \hat{\sigma}_+)$

Transmon-Specific Considerations

- $\delta = (E_{12} - E_{01})/\hbar$
- $\omega_{01}(\Phi_{\text{ext}})$
- $E_c = \frac{e^2}{2C}$
- $E_j \gg E_c$

4 Qubit Control

Readout and the Dispersive Regime

- $\hat{H} = \omega_r \hat{a}^\dagger \hat{a} + \frac{\omega_q}{2} \hat{\sigma}_z - g(\hat{a}^\dagger \hat{\sigma}_- + \hat{a} \hat{\sigma}_+)$
- $\hat{H} \approx (\omega_r + \chi \hat{\sigma}_z) \hat{a}^\dagger \hat{a} + \frac{\omega_q}{2} \hat{\sigma}_z$
- $g \ll |\omega_q - \omega_r|$
- $\gamma_p = \left(\frac{g}{\Delta}\right)^2 \kappa$
- $\kappa = \frac{\omega_r}{Q}$
- $\chi/\kappa \approx 1$
- $\kappa_{\text{eff}}(\omega_q) \ll \kappa(\omega_r)$

Transmon Hamiltonian and Microwave Drive

- $\hat{H} = \frac{\hat{Q}^2}{2C_\Sigma} + \frac{\hat{\Phi}^2}{2L} + \frac{C_d}{C_\Sigma} V_d(t) \hat{Q}$
- $\hat{Q} = -iQ_{\text{zpf}}(\hat{a} - \hat{a}^\dagger)$
- $Q_{\text{zpf}} = \sqrt{\frac{\hbar}{2Z_0}}$
- $Z_0 = \sqrt{\frac{L}{C_\Sigma}}$
- $\hat{a} \rightarrow \hat{\sigma}_-, \hat{a}^\dagger \rightarrow \hat{\sigma}_+$
- $\hat{H} = -\frac{\omega_q}{2} \hat{\sigma}_z + \Omega V_d(t) \hat{\sigma}_y$

Rotating Frame and Rotating Wave Approximation

- $U(t) = e^{i\omega_q t \hat{\sigma}_z / 2}$
- $V_d(t) = V_0 s(t)(I \sin(\omega_d t) - Q \cos(\omega_d t))$
- $\hat{H}'_d = -\frac{\Omega}{2} V_0 s(t)(I \hat{\sigma}_x + Q \hat{\sigma}_y)$
- $\omega_d \approx \omega_q$

Virtual Z Gates and Phase Control

- $I' = I \cos \phi - Q \sin \phi$
- $Q' = I \sin \phi + Q \cos \phi$
- $\phi = \pi/2$

5 Qubit Characterization

Bloch Sphere and State Representation

- $|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$
- $\vec{r} = \begin{pmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{pmatrix}$

Region	Axis Designation	Physical Counterpart
North Pole	$+z$	Longitudinal Axis
South Pole	$-z$	Longitudinal Axis
Equator	x - y Plane	Transverse Plane
Precession	$\phi(t)$	Rotation at ω_{01} (Lab Frame)

Reference Frames and Resonance

- $\omega_r = \omega_{01}$
- $H' = 0$

Decoherence and Relaxation Rates

- $\Gamma_1 \equiv 1/T_1$
- $\Gamma_1 = \Gamma_{1\downarrow} + \Gamma_{1\uparrow}$
- $\Gamma_{1\uparrow} = \Gamma_{1\downarrow}e^{-\hbar\omega/k_B T}$
- Γ_ϕ
- $\Gamma_2 = \frac{\Gamma_1}{2} + \Gamma_\phi$
- $\frac{1}{T_2^*} = \frac{1}{2T_1} + \frac{1}{T_\phi}$

Characterization Measurement Protocols

- $f_q = 3774.16 \text{ MHz}$
- 0.054
- 0.027

Coherence Measurements

- $f(t) \propto e^{-t/T_1}$
- $X_{\pi/2} \text{ — wait } \tau \text{ — } X_{\pi/2}$
- $X_{\pi/2} \text{ — wait } \tau/2 \text{ — } Y_\pi \text{ — wait } \tau/2 \text{ — } X_{\pi/2}$
- $T_2 > T_2^*$

6 TLine

Transmission Line Parameters

- **Resistance per unit length:** R_0 [Ω/m]
- **Inductance per unit length:** L_0 [H/m]
- **Capacitance per unit length:** C_0 [F/m]
- **Conductance per unit length:** G_0 [S/m]

The Telegrapher Equations

- **Voltage equation:**

$$\frac{\partial V(z, t)}{\partial z} = -R_0 i(z, t) - L_0 \frac{\partial i(z, t)}{\partial t}$$

- **Current equation:**

$$\frac{\partial i(z, t)}{\partial z} = -G_0 V(z, t) - C_0 \frac{\partial V(z, t)}{\partial t}$$

- **Lossless line:** for $R_0 = 0$ and $G_0 = 0$,

$$\frac{\partial^2 V}{\partial z^2} = L_0 C_0 \frac{\partial^2 V}{\partial t^2}$$

Wave Propagation Dynamics

- **Propagation speed:**

$$v = \frac{1}{\sqrt{L_0 C_0}}$$

- **Voltage solution:**

$$V(z, t) = V^+(t - z/v) + V^-(t + z/v)$$

- **Characteristic impedance:**

$$Z_0 = \sqrt{\frac{L_0}{C_0}}$$

- **Current waves:**

$$i^+(z, t) = \frac{V^+(z, t)}{Z_0}, \quad i^-(z, t) = -\frac{V^-(z, t)}{Z_0}$$

Reflections and Terminations

- **Voltage reflection coefficient:**

$$\rho_v = \Gamma_L = \frac{R_L - Z_0}{R_L + Z_0}$$

- **Current reflection coefficient:**

$$\rho_i = -\rho_v = \frac{Z_0 - R_L}{Z_0 + R_L}$$

- **Matched load:** if $R_L = Z_0$, then $\Gamma_L = 0$ and no reflections occur.

Power on the Transmission Line

- **Instantaneous power:**

$$p(z, t) = V(z, t) i(z, t)$$

- **Average power for a traveling wave:**

$$P_{\text{avg}} = \frac{V_{\text{rms}}^2}{Z_0} = I_{\text{rms}}^2 Z_0$$

- **Power transfer:** maximum power is delivered when the line is matched to the load.

7 Smatrix

Scattering Parameters for an Impedance Discontinuity

- **Impedances:** Two semi-infinite lines meet at a junction with characteristic impedances Z_1 and Z_2 .

- **Power-normalized waves:**

$$a_n = \frac{V_n^+}{\sqrt{Z_n}}, \quad b_n = \frac{V_n^-}{\sqrt{Z_n}}$$

- **Port convention:** Port 1 corresponds to Z_1 , and Port 2 corresponds to Z_2 .

Reflection and Transmission Coefficients

- **Reflection at the junction from Port 2:**

$$S_{22} = \Gamma_{A+} = \frac{Z_1 - Z_2}{Z_1 + Z_2}$$

- **Voltage transmission coefficient:**

$$\frac{V_A^-}{V_A^+} = \frac{2Z_1}{Z_1 + Z_2}$$

- **Power-normalized transmission:**

$$S_{12} = \sqrt{\frac{Z_2}{Z_1}} \cdot \frac{V_A^-}{V_A^+} = \frac{2\sqrt{Z_1 Z_2}}{Z_1 + Z_2}$$

- **Symmetric result:** by reciprocity,

$$S_{21} = \frac{2\sqrt{Z_1 Z_2}}{Z_1 + Z_2}$$

- **Reflection seen from Port 1:**

$$S_{11} = \frac{Z_2 - Z_1}{Z_1 + Z_2}$$

Scattering Matrix

$$\mathbf{S} = \frac{1}{Z_1 + Z_2} \begin{pmatrix} Z_2 - Z_1 & 2\sqrt{Z_1 Z_2} \\ 2\sqrt{Z_1 Z_2} & Z_1 - Z_2 \end{pmatrix}$$

Device Properties

- **Reciprocal:** $S_{12} = S_{21}$.
- **Not symmetric:** generally $S_{11} \neq S_{22}$ unless $Z_1 = Z_2$.
- **Lossless junction:** the S-matrix is unitary, $\mathbf{S}\mathbf{S}^\dagger = \mathbf{I}$.

Fundamental Power Wave Definitions

- **Incident wave:**

$$a_i = \frac{V_i + Z_{ri}I_i}{2\sqrt{Z_{ri}}}$$

- **Reflected wave:**

$$b_i = \frac{V_i - Z_{ri}I_i}{2\sqrt{Z_{ri}}}$$

- **Power normalization:** $|a_i|^2$ and $|b_i|^2$ represent power in watts.

- **Net real power at port i :**

$$P_i = |a_i|^2 - |b_i|^2$$

- **Spatial propagation:**

$$a(x) = a(0)e^{-jkx}, \quad b(x) = b(0)e^{+jkx}$$

- **Reference impedance:** Z_{ri} is the characteristic impedance used to normalize port waves.

General N-Port S-Matrix Framework

- **Black-box model:** internal circuit details are replaced by input-output relations between incident and reflected waves.

- **S-matrix relation:**

$$\mathbf{b} = \mathbf{S}\mathbf{a}$$

- **Matrix size:** for an N -port network, $\mathbf{S}$ is an $N \times N$ matrix.

- **Scattering parameter definition:**

$$S_{ij} = \left. \frac{b_i}{a_j} \right|_{a_k=0 \text{ for all } k \neq j}$$

- **Matched-load condition:** $a_k = 0$ is realized by terminating all non-driven ports with their reference impedances.

Two-Port Network

- **Equations:**

$$b_1 = S_{11}a_1 + S_{12}a_2, \quad b_2 = S_{21}a_1 + S_{22}a_2$$

- **Matrix form:**

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

Directional Coupler Performance

- **Coupling factor (dB):**

$$\text{Coupling (dB)} = -10 \log \left(\frac{P_{\text{coupling forward}}}{P_{\text{incident}}} \right)$$

- **Isolation factor (dB):**

$$\text{Isolation (dB)} = -10 \log \left(\frac{P_{\text{coupled reverse}}}{P_{\text{incident}}} \right)$$

- **Directivity (dB):**

$$\text{Directivity (dB)} = \text{Isolation (dB)} - \text{Coupling (dB)}$$

- **Mainline loss:** total insertion loss in the primary through-line.

S-Parameter Extractions

- **Forward measurements:** when SW1 is in position 1,

$$S_{11} = \frac{b_1}{a_1}, \quad S_{21} = \frac{b_2}{a_1}$$

- **Reverse measurements:** when SW1 is in position 2,

$$S_{22} = \frac{b_2}{a_2}, \quad S_{12} = \frac{b_1}{a_2}$$

- **2×2 S-matrix:**

$$\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$$

- **Index convention:** i = destination/output port, j = source/input port.

8 RF Components and VNA Fundamentals

Coupling Factor (dB)

- Measures the power diverted to the coupled path relative to the incident power.

$$\text{Coupling Factor (dB)} = -10 \log \left(\frac{P_{\text{coupling forward}}}{P_{\text{incident}}} \right)$$

VNA Signal Measurements

- RX_{REF} : measures a_1 (incident wave)
- RX_{TEST1} : measures b_1 (reflected wave)
- RX_{TEST2} : measures b_2 (transmitted wave)

9 Complex Signal Foundations

Euler's Formula

- The mathematical basis for representing real signals in the complex domain.

$$e^{j\omega t} = \cos \omega t + j \sin \omega t$$

$$e^{-j\omega t} = \cos \omega t - j \sin \omega t$$

Trigonometric to Complex Exponential Conversion

- Converting sinusoids to exponential forms:

$$2 \cos \omega t = e^{j\omega t} + e^{-j\omega t}$$

$$2j \sin \omega t = e^{j\omega t} - e^{-j\omega t}$$

Magnitude and Phase Extraction

- Magnitude line spectra:

$$2\sqrt{A^2(f) + B^2(f)}$$

- Phase spectra:

$$\tan^{-1} \left(\frac{B(f)}{A(f)} \right)$$

10 Signal Modulation

General Sine Wave Equation

- **Time-domain representation:**

$$v(t) = A \cdot \sin(2\pi ft + \phi)$$

where A is amplitude, f is frequency, and ϕ is phase.

Amplitude Modulation (AM)

- **Time-domain equation:**

$$s(t) = [A_c + A_m \cos 2\pi f_m t] \cos 2\pi f_c t$$

where A_c is carrier amplitude, A_m is modulation amplitude, f_c is carrier frequency, and f_m is modulation frequency.

- **Trigonometric identity for frequency shifting:**

$$\cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)]$$

- **Expanded AM equation (sidebands):**

$$s(t) = A_c \cos 2\pi f_c t + \frac{A_m}{2} \cos 2\pi(f_c + f_m)t + \frac{A_m}{2} \cos 2\pi(f_c - f_m)t$$

11 Quadrature (I/Q) Modulation and Demodulation

I/Q Modulation Time-Domain Equation

- **General form:**

$$y(t) = g_1(t) \cos 2\pi f_c t - g_2(t) \sin 2\pi f_c t$$

where $g_1(t)$ is the In-Phase (I) signal and $g_2(t)$ is the Quadrature (Q) signal.

Complex Baseband Signal Representation

- **Complex envelope:**

$$g(t) = g_1(t) + jg_2(t)$$

- **Quadrature modulation (complex notation):**

$$v(t) = \text{Re} \{ g(t) e^{j2\pi f_c t} \}$$

- **Expanded form:**

$$v(t) = g_1(t) \cos(2\pi f_c t) - g_2(t) \sin(2\pi f_c t)$$

Quadrature Demodulation Process

- **Total received signal ($x(t)$):**

$$x(t) = y(t) e^{-j2\pi f_c t}$$

- **In-Phase extraction ($z_1(t)$):**

$$z_1(t) = \frac{1}{2} g_1(t) + [\text{high-frequency terms}]$$

- **Quadrature extraction ($z_2(t)$):**

$$z_2(t) = \frac{1}{2} g_2(t) + [\text{high-frequency terms}]$$

(after low-pass filtering to remove high-frequency components)

Signal Examples

- **Sum of three tones (real):**

$$s_1(t) = 10 \cos 2\pi 100t + \cos 2\pi 200t + 4 \cos 2\pi 300t$$

- **Sum of three tones with phase shift:**

$$s_2(t) = 10 \cos \left(2\pi 100t + \frac{\pi}{4} \right) + \cos \left(2\pi 200t + \frac{\pi}{6} \right) + 4 \cos 2\pi 300t$$

12 Digital Signal Processing

Sampling and Aliasing

- **Sampling Rate (f_s):** The inverse of the sampling period (t_s).

$$f_s = \frac{1}{t_s}$$

- **Nyquist-Shannon Criterion (Undersampling):** For a band-limited signal between frequencies f_l and f_h .

$$f_s > 2(f_h - f_l)$$

- **Alias Frequency:** The frequency at which a signal f appears when sampled at f_s .

$$f_{\text{alias}} = |f - n \cdot f_s|$$

- **Spectral Image Locations (DAC):** Positions of unwanted images after D/A conversion.

$$\text{Location} = N \cdot f_s \pm f$$

where N is an integer and f is the generated signal frequency.

Quantization and Performance Metrics

- **Quantization Step (q):** The voltage difference between two adjacent discrete levels.

$$q = \frac{V_{\text{FS}}}{2^N - 1}$$

where $V_{\text{FS}} = 2V_{\text{max}}$ is the full-scale voltage.

- **ADC Resolution (Δ):**

$$\Delta = \frac{V_{\text{max}} - V_{\text{min}}}{2^n}$$

- **Dynamic Range (DR):** Logarithmic ratio between the largest and smallest measurable values.

$$\text{DR} = 20 \log_{10}(2^N) \approx 6.02N \text{ [dB]}$$

- **Quantization Noise:** Idealized model of the noise characteristics.

$$\text{Noise Power} = \frac{q^2}{12}$$

- **Spurious Free Dynamic Range (SFDR):**

$$\text{SFDR} = \frac{P_{\text{fundamental}}}{P_{\text{biggest_spur}}}$$

Fixed-Point Data Representation (Qn.b format)

- **Unsigned Range:**

- Minimum: 0
- Maximum: $2^n - 2^{-b}$

- **Signed 2's Complement Range:**

- Minimum: -2^{n-1}
- Maximum: $2^{n-1} - 2^{-b}$

- **Precision/Resolution:** The smallest representable number.

$$\text{Resolution} = 2^{-b}$$

- **Arithmetic Bit-Width Expansion:**

- Addition/Subtraction: Requires $n + 1$ bits to avoid overflow.
- Multiplication: Multiplying $Q(m1.n1)$ and $Q(m2.n2)$ results in $Q(m1 + m2).(n1 + n2)$.

Multirate Operations

- **Decimation:** Reducing the sampling rate by an integer factor M .

$$f_d = \frac{f_s}{M}$$

- **Interpolation:** Increasing the sampling rate by an integer factor L .

$$f_u = L \cdot f_s$$

- **Interpolated Image Location:** The location of the first image after interpolation by factor K .

$$f_{\text{image}} = K \cdot f_c - f_o$$

Data Converter Hardware Dynamics

- **Interleaved ADC Sampling Phase:** The phase offset for the n^{th} sub-ADC in a system of m interleaved converters.

$$\theta_n = \frac{2\pi(n-1)}{m}$$

- **DAC Frequency Response (Normal Mode):** Follows a sinc envelope.

$$H(f) = \frac{\sin(x)}{x}, \quad \text{where } x = \frac{\pi f}{f_s}$$

- **DAC Frequency Response (Mix-Mode):** Reshapes energy for the 2nd Nyquist Zone.

$$H(f) = \frac{\sin^2(x)}{x}, \quad \text{where } x = \frac{\pi f}{2f_s}$$

13 Semiconductor Qubits and the Elzerman Method

Energy Level and Temperature Requirements

- **Zeeman Energy (ΔE_Z):** Approximately $200 \mu\text{eV}$ at a magnetic field of 10 T.
- **Thermal Energy (kT):** Approximately $25 \mu\text{eV}$ at a temperature of 0.3 K.
- **Orbital Energy Spacing ($\Delta E_{\text{orbital}}$):** Approximately 1.1 meV.
- **System Requirement Formula:** To ensure a valid qubit state, the parameters must satisfy:

$$\Delta E_{\text{orbital}} \gg \Delta E_Z \gg kT$$

Tuning and Bias Formulas

- **Optimal Bias Point (V_M):** The voltage V_M sets the bias point for energy level oscillations.
- **Optimal Working Point Calculation:** If the crossing point where the spin-up electron energy exceeds the Fermi level is V_{M0} , the optimal working point to place the Fermi energy halfway between levels is:

$$V_M = V_{M0} + \frac{1}{2e} \Delta E_Z$$

Measurement Fidelity and Probability

- **Error Probability α :** The probability of measuring “down” when the state is spin-up.

$$\alpha = P(\text{“down”} \mid \text{spin-} \uparrow)$$

- **Error Probability β :** The probability of measuring “up” when the state is spin-down.

$$\beta = P(\text{“up”} \mid \text{spin-} \downarrow)$$

- **Decay and Detection Rate (β_1):**

$$\beta_1 = \frac{1}{1 + T_1 \Gamma_{\downarrow}}$$

- **Combined β Estimation:**

$$1 - \beta \sim (1 - \beta_1)(1 - \beta_2) + \alpha\beta_1$$

- **Spin-up Fidelity:** $1 - \alpha$

- **Visibility Formula:**

$$\text{Visibility} = 1 - \alpha - \beta$$

Qubit Interaction and Control

- **Hamiltonian Basis:** The baseline Hamiltonian for the split levels is represented as:

$$H_0 \sim \sum \omega_i \sigma_{zi}$$

- **Exchange Interaction (J):** In two-qubit systems, the exchange interaction is computed using the difference in resonance frequencies (f) of one qubit based on the state of the other:

$$J = f_2 - f_1 \quad \text{or} \quad J = f_4 - f_3$$

where f_1/f_2 are resonance frequencies for q_0 given q_1 states, and f_3/f_4 are for q_1 given q_0 states.

Randomized Benchmarking (RB)

- **Data Fitting:** Success probabilities $p(m)$ for a sequence of m gates are processed using:

$p(m)$ fitted to a linear combination of exponential decays

- **SPAM Fidelity Estimation:** State Preparation and Measurement (SPAM) error can be estimated by evaluating the least-squares fit of the RB decay curve at:

$$m = 0$$